SF2 SEMANTIC FOLDING SENSOR FUSION

SF2 CASE STUDY

SF2 Systems for Intelligent Water Pump Monitoring

Failure detected 4 days before breakdown

Trained on 220K records in under 15 minutes – on a standard notebook.

52 Sensors monitoring the pump

220,000+ Data points processed

<15 min Training on standard notebook

4 Days Advance failure dectection

THE CHALLENGE – Why Traditional Monitoring AI Fails

A municipal water pump dataset with many measurements but very few failures making it difficult to train with traditional AI:

- **Only 7 failure events (<1%) in 220k records** → not enough for ML
- **No sensor metadata (types, units unknown)** → no feature engineering possible
- **High-dimensional sensor interactions** → impossible to model manually
- **Traditional AI requires large labeled datasets, long training cycles and high compute resources**

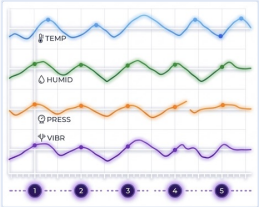
Threshold systems only detect when something has already gone wrong. SF2 detects degradation trends, not just sudden failures.

> The problem is not data availability, it's understanding system behavior.

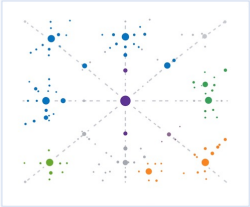
THE SF2 SOLUTION – Decoding the Language of Complex Systems

SF2 converts entire system states into a unique digital fingerprint – essentially a FaceID for machines – that encapsulates the full context and interdependencies of a system at any given moment.

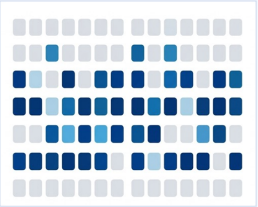
Raw Sensor Data



SF2 Sensor Fusion

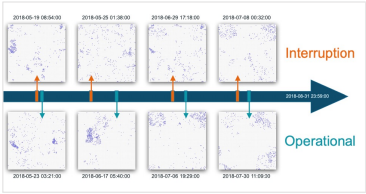


System State Fingerprint

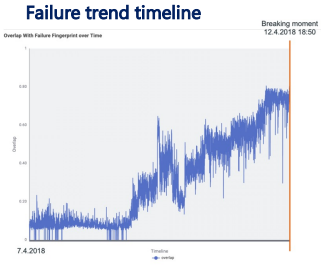


THE RESULT – Predictive Maintenance in Minutes, Not Months

Operational system state versus failure



Failure trend timeline



Proof of Performance

- ✓ Failure trajectory detected **4 days in advance**
- ✓ **Reliable results even with minimal failure data**
- ✓ **Trained** on standard hardware in **~15 minutes**
- ✓ **No system or data science knowledge required**

SF2 makes complex systems understandable — and failures predictable.

HOW SF2 WORKS

1 Data Normalization (Binning)

Raw time-series sensor values are converted into discrete standardized categories (e.g., 255 bins per sensor) to prepare for semantic mapping.

2 Semantic Mapping (Library)

Using only normal operating data, the system learns baseline behavior & natural relationships between sensors, creating a model of healthy operation.

3 Fingerprint Aggregation (Stacking)

Each sensor value is mapped to a sparse binary vector. These are combined into a single system-state fingerprint representing the entire asset.

4 Real-Time Interpretation (Comparison)

New fingerprints are compared against known patterns to detect deviations and predict trends before failures occur.

Multiple sensor fingerprints are stacked and aggregated into a single representation.
Each timestamp becomes a context that captures the entire system state at that moment.

Why Understanding System State Outperforms Statistical Estimations

SF2 (Deterministic)

- ✓ Accurate state-based monitoring & prediction
- ✓ Detects gradual shifts as they occur
- ✓ No labelled date required
- ✓ Explainable and reproducible
- ✓ Real-time, low compute
- ✓ Data sovereignty (100% local)
- ✓ Edge-ready – runs in real-time on any device

Traditional AI (Probabilistic)

- × Uncertain approximations
- × Detects failures only after thresholds are crossed
- × Requires large labeled datasets
- × Black-box behavior
- × High compute & high-latency processing
- × Mostly cloud-dependent (data & security risk)
- × Not edge-ready

MEASUREABLE BUSINESS IMPACT

Less Downtime
Avoid costly failures before they occur

Faster Time-to-Value
Deploy in days, not months

Predictable Operations
Plan maintenance with confidence

Lower Cost
No heavy infrastructure required

Full Control
100% local, secure and transparent

SF2 makes system states directly comparable in real time. Deviations from normal operation become visible early, enabling **planned maintenance** and **reducing unplanned downtime**. This is achieved without predefined thresholds or model retraining. As a result, system behavior remains interpretable even under changing operating conditions. All analysis is **performed locally** and does not require **no prior knowledge** of the physical system, sensor types or failure patterns.

CONCLUSION

SF2 turns noisy sensor steams into a unified live representation of system behavior.
In this case study, signs of failure in a municipal waterpump were visible 4 days before breakdown, after training with only 220K+ records and 7 failure events in under 15 minutes on a standard hardware.

Learn more at www.SF2systems.com

FUTURE POTENTIAL – Any system with sensors can be understood

Manufacturing	Chemical & Process	Pharmaceutical	Industrial Equipment
Energy Systems	Water & Utilities	Oil & Gas	Smart Infrastructure
Smart Buildings	Automotive Systems	Aerospace & Aviation	Mobility & Transportation

SF2 SYSTEMS

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